HW 5

Monday, 13 February 2023 4:58 PM

Problem 12.1

$$\hat{r}(c) = \frac{k \cdot c}{K' + Kc + c^2}$$

(i) Max reaction rate:

$$\frac{d\hat{r}}{dc} = \frac{(K' + Kc + c^2)\cdot k - k\cdot c\cdot(K + 2c)}{(K' + Kc + c^2)^2} = 0$$

$$\Rightarrow \frac{kK' + kKc + kc^2 - kKc - 2kc^2}{(K' + Kc + c^2)^2} = 0$$

$$\Rightarrow \frac{kK' - kc^2}{(K' + Kc + c^2)^2} = 0$$

$$kK' = kc^2$$

$$c = \sqrt{K'} \longrightarrow \text{Point of Local extrema [Taking positive value)}$$

Max? $\frac{d^2\hat{r}}{dc^2} = \frac{(K' + Kc + c^2)^{\cancel{2}}\cdot(-2kc) - 2(K' + Kc + c^2)\cdot(K + 2c)\cdot(kK' - kc^2)}{(K' + Kc + c^2)^{\cancel{4}3}}$; @ maximum critical point: $\frac{d^2\hat{r}}{dc^2} < 0$

$$= \frac{(K' + Kc + c^2)(-2kc) - 2(K + 2c)(kK' - kc^2)}{(K' + Kc + c^2)^3}$$

$$= \frac{-2kK'c - 2kKc^2 - 2kc^3 - 2kKK' + 2kKc^2 - 4kK'c + 4kc^3}{(K' + Kc + c^2)^3}$$

$$\frac{d^2\hat{r}}{dc^2} = \frac{2kc^2 - 6kK'c - 2kKK'}{(K' + Kc + c^2)^3}$$

@ $c = \sqrt{K'} \Rightarrow \frac{2kK' - 6kK'\sqrt{K'} - 2kKK'}{(K' + \sqrt{K}K + K')^3}$

K', k, K are defined as positive constants :→ $(K' + K\sqrt{K'} + K)^3 > 0$ [Denominator is always positive]

Numerator: $kK'\underbrace{(2 - 6\sqrt{K'} - 2K)}$

This has to be a negative value since K and K' cannot be less than zero

(ii) Plate type catalyst pellet (half thickness a): $u = \frac{c}{c_s}$; $c = u\cdot c_s$; $\hat{r}(c_s) = \frac{kc_s}{K' + Kc_s + c_s^2}$

$$R(u) = \frac{\hat{r}(c)}{\hat{r}(c_s)} = \frac{\frac{\cancel{k}c}{K' + Kc + c^2}}{\frac{\cancel{k}c_s}{K' + Kc_s + c_s^2}} = \frac{(c/c_s)}{(K' + Kc + c^2)}\cdot(K' + Kc_s + c_s^2)$$

$$= \frac{(c/c_s)\qquad (K' + Kc_s + c_s^2)}{\left[\frac{K'}{K' + Kc_s + c_s^2} + \frac{Kc}{K' + Kc_s + c_s^2} + \frac{c^2}{K' + Kc_s + c_s^2}\right](K' + Kc_s + c_s^2)}$$

↓ Taking $(K' + Kc_s + c_s^2)$ as common factor from denominator

↓ $u = \frac{c}{c_s}$

$$= \frac{u}{\left[\underbrace{\frac{K'}{K' + Kc_s + c_s^2}}_{\alpha} + \underbrace{\frac{Kc_s u}{K' + Kc_s + c_s^2}}_{\beta} + \underbrace{\frac{u^2 c_s^2}{K' + Kc_s + c_s^2}}_{\gamma}\right]}$$

$$\therefore \boxed{R(u) = \frac{u}{\alpha + \beta u + \gamma u^2}}$$

(ii) Non-linear kinetics $\quad \frac{d^2u}{d\rho^2} = \phi^2 R(u) \quad , \quad 0<\gamma<1 \quad , \quad \left.\frac{du}{d\rho}\right|_{\rho} = 0 \quad , \quad u(1)=0$

$$\phi^2 = \frac{\hat{r}(c_s,T)\rho_b S_g a^2}{De\, c_s} \quad ; \quad \eta = \frac{1}{\phi^2}\left.\frac{du}{d\rho}\right|_{\rho=1}$$

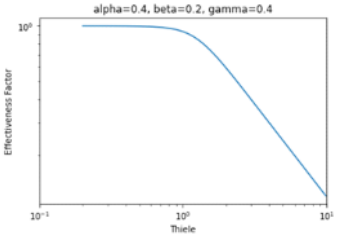
As shown in lecture, we define $v(\rho) = \frac{du}{d\rho}$; $F(u) = 2\int_0^u R(u)\cdot du$

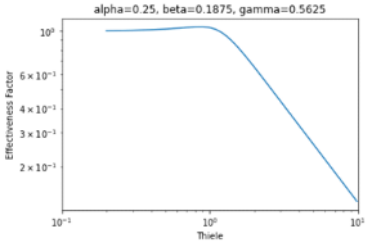
$$\phi = \int_{u_0}^{1} \frac{du}{\sqrt{F(u)-F(u_0)}} \rightarrow \text{Solve for } u_0 \text{ numerically}$$

Then use u_0 to solve: $\eta = \frac{v(1)}{\phi^2} = \frac{1}{\phi}\sqrt{F(1)-F(u_0)}$

Solving this system in python gives the following graphs:

(iii) (iv)





(v) Differences: For $[\alpha = 0.25 \; ; \; \beta = 0.1875 \; ; \; \gamma = 0.5625]$ the effectiveness factor goes above 1. Physically this means that the diffusion rate increases more than avg. reaction rate (if the rate of reaction is evaluated at surface) (given the higher alpha value but lower gamma value)

Problem 12.3

Plate catalyst pellet; First-order reaction : (w/ no external mass transfer limitations)

$$\eta = \frac{\tanh\phi}{\phi}$$

Regular perturbation theory: $\eta = 1 - \kappa_1\phi^2 + \kappa_2\phi^4 + O(\phi^6)$; $-\kappa_1 = -\frac{1}{V^*}\int_{V^*} \|\nabla u_1\|^2 dV^*$

$$\kappa_2 = \frac{1}{V^*}\int_{V^*} u_1^2\, dV^*$$

For plate of half-Thickness a : $\kappa_1 = \frac{1}{3}$

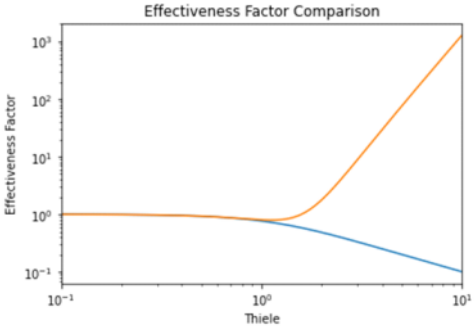
$\kappa_2 = \frac{2}{15}$

The plots below show the graphs for (i) & (ii) :

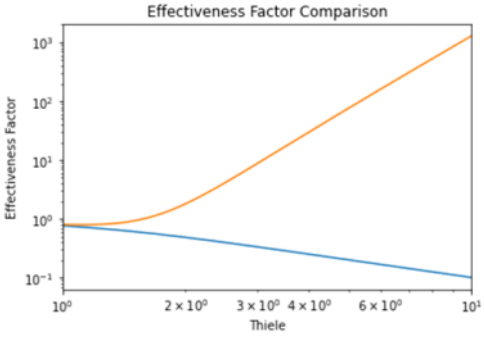
The regular perturbation is only accurate for values between $0 < \phi < 1$ as shown by the graph here;

However, the effectiveness factor approximation blows up for large ϕ values.

$0 < \phi < 1$



(iii) Uniform pore size distribution in plate catalyst is same as that in part (i)



$1 < \phi < 10$